A Geometric Framework for Understanding Memorization in Generative Models

Brendan Leigh Ross, Hamidreza Kamkari, Zhaoyan Liu, Tongzi Wu, George Stein, Gabriel Loaiza-Ganem, Jesse C. Cresswell

Layer 6 AI, Toronto, Canada

Background [1]

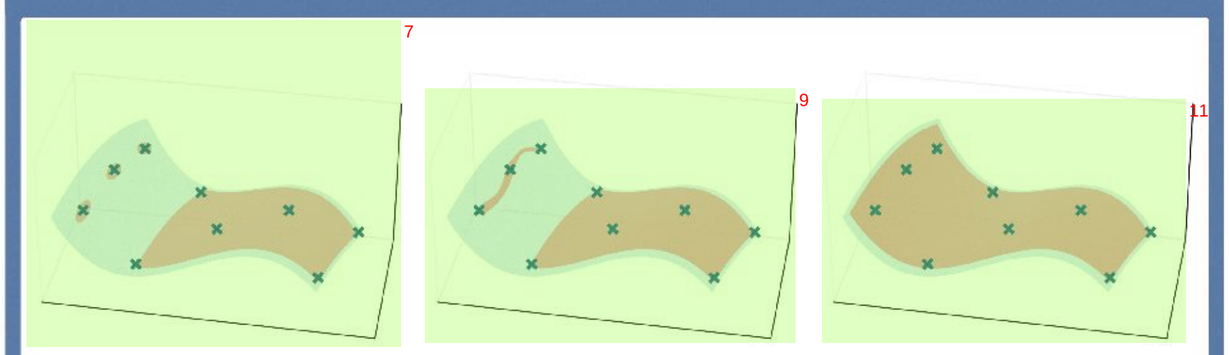
Memorization and Overfitting in Deep Generative Models [2]

- Deep generative models are known to memorize and reproduce their training data [3]
- This behaviour reduces model utility and may have severe privacy and legal implications
- Contributing factors [4]: data complexity, duplication of training points, and highly specific conditioning
- There is as yet no systematic framework describing memorization in DGMs

The Manifold Hypothesis [4]

- High-dimensional data lies on a manifold $\mathcal{M}_*$ with multiple components of varying dimensionalities [5]
- DGMs also generate samples from a manifold $\mathcal{M}_\theta$ with the goal of matching $\mathcal{M}_\theta$ to $\mathcal{M}_*$
- The local intrinsic dimension of $x \in \mathcal{M}_*$ (resp. $x \in \mathcal{M}_\theta$), denoted $\text{LID}_*(x)$ (resp. $\text{LID}_\theta(x)$), is the dimension of the manifold component containing x
- Intuitively, the LID of x captures the complexity of the datapoint x and constraints on the dataset's structure

Our Geometric Framework [6]



(a) Exact memorization [8] ($\text{LID}_\theta(x) = 0$) (b) Some Memorization [10] ($\text{LID}_\theta(x) = 1$) (c) No memorization [12] ($\text{LID}_\theta(x) = 2$)

Figure 1: LID values for three red model manifolds with different degrees of memorization of the three leftmost datapoints, which are depicted as dark blue crosses. The true manifold is light blue. [13]

We propose that $x \in \mathcal{M}_*$ is memorized when $\text{LID}_\theta(x)$ is small. [14]

- This encompasses both exact copying of training images ($\text{LID}_\theta(x) = 0$) and copying of styles or parts of an image as in Figure 2 ($\text{LID}_\theta(x) > 0$ but still small) [15]

From this definition, memorization can be split into two types: [16]

1. $\text{LID}_\theta(x) < \text{LID}_*(x)$: *overfitting-driven memorization* (OD-Mem), in which the model fails to generalize to the ground truth manifold correctly (Figure 1) [17]
2. $\text{LID}_\theta(x) = \text{LID}_*(x)$: *data-driven memorization* (DD-Mem), in which the ground truth data distribution has been learned correctly, but itself contains very few degrees of freedom at x

Figure 2: 8 images along a relatively low-dimensional manifold learned by Stable Diffusion v1.5. The first is a real image from LAION (flagged as memorized by [2]), and the remainder were generated by the model. [19]

Explaining Memorization [20]

The geometric framework explains memorization-related phenomena in the literature: [21]

Duplicated Data It has been observed that memorization occurs when training points are duplicated. Here we identify that data duplication is equivalent to an LID of 0: [22]

Theorem (Informal). With high probability and a sufficiently large dataset, duplicate samples occur at x_0 if and only if $\text{LID}_*(x_0) = 0$. [23]

Complexity. [4] observes that simple images are memorized more often. This is neatly explained by the fact that they have lower LID_*. [24]

Conditioning. Highly-specific conditioning inputs encourage the reproduction of memorized samples. We explain this by the fact that conditioning on a prompt c decreases LID: [25]

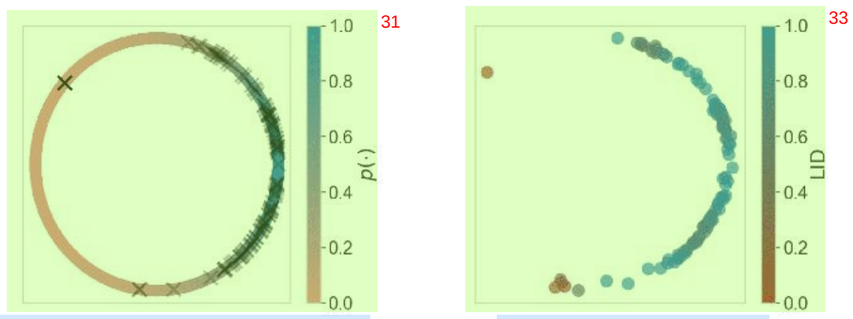
Proposition. $\text{LID}_\theta(x|c) \leq \text{LID}_\theta(x)$. [26]

Classifier-Free Guidance (CFG). A large classifier-free guidance vector in diffusion models correlates strongly with memorization [5]. This is explainable by observations in the literature that faster explosion of model norms during sampling corresponds to lower LID_θ. [27]

Empirical Verification [28]

We can test this framework using the LID_θ estimator of [3]: [29]

von Mises Distribution We train a diffusion model on a von Mises distribution; in Figure 3b we depict model samples with their corresponding LIDs, and a couple of samples are clearly memorized with $\text{LID}_\theta(x) = 0$, while the remaining samples have $\text{LID}_\theta(x) = 1$. [30]



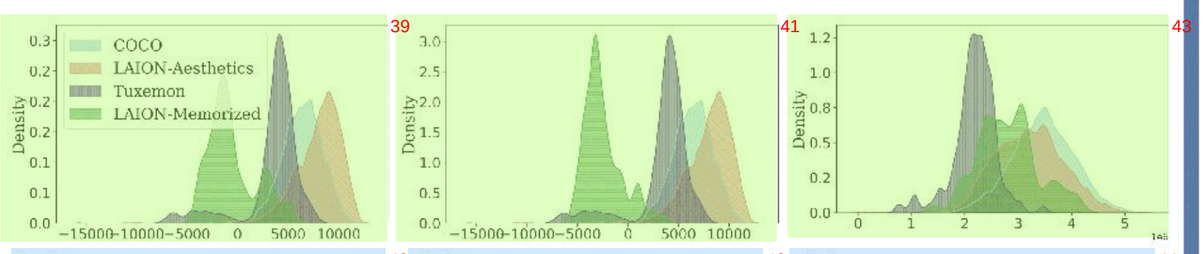
(a) Ground truth manifold. [32] (b) Model samples. [34]

Figure: Training a diffusion model on a von Mises distribution. [35]

Stable Diffusion on LAION, COCO, and Tuxemon We compute LID_θ of Stable Diffusion [1] on [36]

- Unmemorized images from a mix of LAION Aesthetics 6.5+, COCO, and Tuxemon [37]
- Memorized images from LAION [2]

We find that low LIDs correspond better to memorized images than PNG size, a proxy for image complexity. [38]



(a) Unconditional LID (AUROC = 97.6%). [40] (b) Conditional LID (AUROC = 99.2%). [42] (c) PNG compression (AUROC = 70.3%). [44]

Figure 4: Density histograms for each memorization metric on the different datasets. [45]

[1] Rombach, Blattmann, Lorenz, Esser, Ommer. "High-resolution image synthesis with latent diffusion models". CVPR 2022.
[2] Webster. A reproducible extraction of training images from diffusion models. arXiv 2305.08694.
[3] Kamkari, Ross, Hosseinzadeh, Cresswell, Loaiza-Ganem. "A geometric view of data complexity: Efficient local intrinsic dimension estimation with diffusion models". arXiv 2406.03537.
[4] Somepalli, Singla, Goldblum, Geiping, Goldstein "Understanding and mitigating copying in diffusion models". NeurIPS 2023.
[5] Wen, Liu, Chen, Lyu "Detecting, explaining, and mitigating memorization in diffusion models". ICML 2023.